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Core Workloads

Pod, Deployment, StatefulSet, DaemonSet, Job, CronJob

Java Backend Architecture Interview Series • Tutorial Edition • Easy Diagrams & Examples

What are Workload Resources?

A Pod is just a wrapper for one or more containers. But Pods are mortal — they die and don't come back. Workload resources (Deployment, StatefulSet, etc.) are 'managers' that ensure the right number of Pods are always running.

■ **Simple Analogy:** A Pod is like one worker. A Deployment is like a staffing agency that always keeps 3 workers on the job — if one quits, the agency hires a replacement immediately.

Workload Types at a Glance

Workload	Use Case	Key Feature	Example
Deployment	Stateless apps	Rolling updates, rollback	REST API, web server
StatefulSet	Stateful apps	Stable pod names + PVC per pod	PostgreSQL, Kafka, Redis
DaemonSet	Run on every node	One pod per node, auto-adds to new nodes	Log agent, monitoring
Job	One-time batch task	Runs to completion, then done	DB migration, report
CronJob	Scheduled batch task	Cron schedule syntax	Nightly cleanup, backup

Deployment — The Most Common Workload

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-api
spec:
  replicas: 3 # run 3 pods
  selector:
    matchLabels:
      app: java-api
  strategy:
    type: RollingUpdate
  rollingUpdate:
    maxUnavailable: 1 # at most 1 pod down during update
    maxSurge: 1 # at most 1 extra pod during update
  template:
    spec:
      containers:
        - name: java-api
          image: myregistry/java-api:v2
          resources:
            requests: { cpu: 250m, memory: 256Mi }
            limits: { cpu: 500m, memory: 512Mi }
```

StatefulSet — For Databases

Unlike Deployment, StatefulSet gives each Pod a STABLE name (pod-0, pod-1, pod-2) and its OWN persistent volume. Pods are created in order and deleted in reverse. Essential for databases that need stable network identity.

```
# StatefulSet pods get stable names:
postgres-0 (primary / leader)
postgres-1 (replica)
postgres-2 (replica)

# Each gets its own PVC:
data-postgres-0 (mounted only by postgres-0)
data-postgres-1 (mounted only by postgres-1)
```

DaemonSet — One Pod Per Node

When you add a new node to the cluster, DaemonSet automatically schedules a pod on it. Perfect for infrastructure agents like log collectors (Fluentd), monitoring (Prometheus Node Exporter), or security scanners.

Job and CronJob

```
# CronJob — run DB cleanup every night at 2 AM
apiVersion: batch/v1
kind: CronJob
metadata:
  name: db-cleanup
spec:
  schedule: "0 2 * * *" # cron syntax
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: cleanup
              image: myregistry/cleanup-job:latest
              restartPolicy: OnFailure
```

■ **Real-World Example:** Spotify runs Kafka on K8s using StatefulSets — each broker gets a stable identity (kafka-0, kafka-1, kafka-2) and its own 2TB EBS volume. They use DaemonSets for Filebeat on every node to ship logs to Elasticsearch.